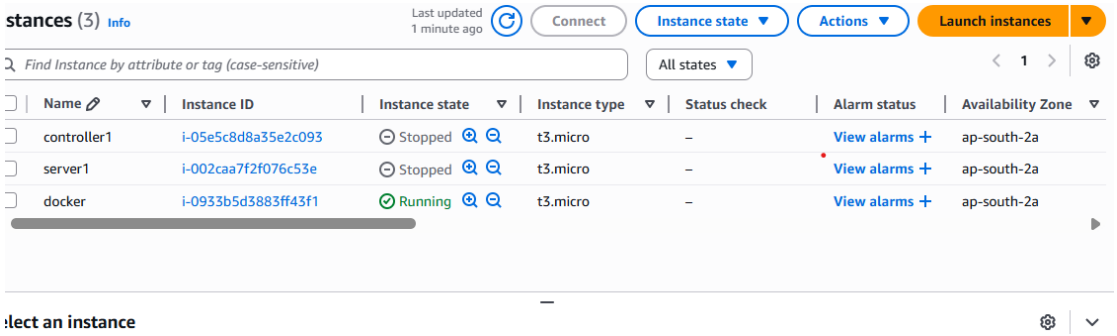
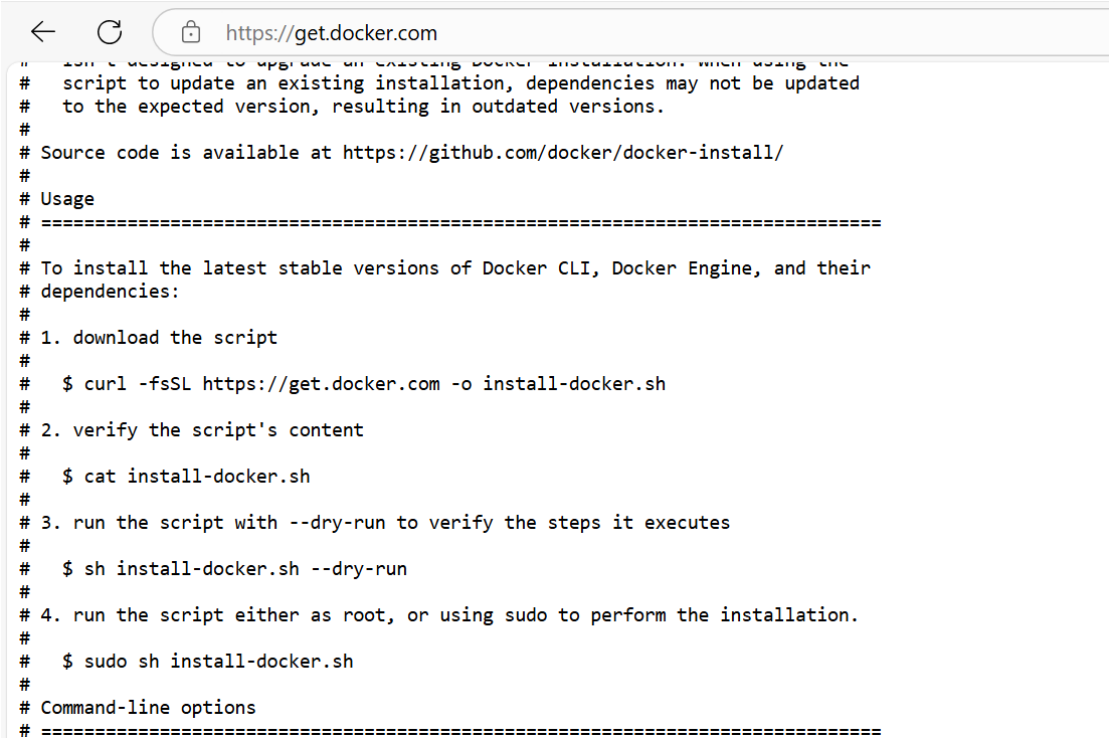
PRACTICAL LAB :

Create an instance

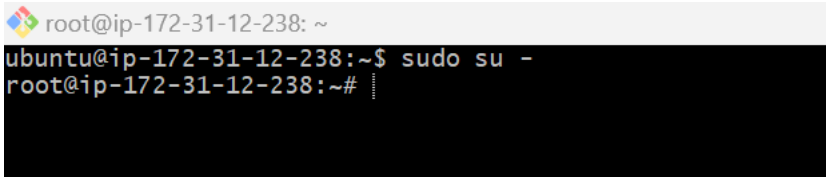
Connect to AWS Create new machine



Open browser (get.docker.com)



Always docker run on root user (#)
sudo su -



Download and install docker

curl -fsSL https://get.docker.com -o install-docker.sh

```
ubuntu@ip-172-31-15-92:~$ sudo su -
root@ip-172-31-15-92:~# curl -fsSL https://get.docker.com -o install-docker.sh
root@ip-172-31-15-92:~#
```

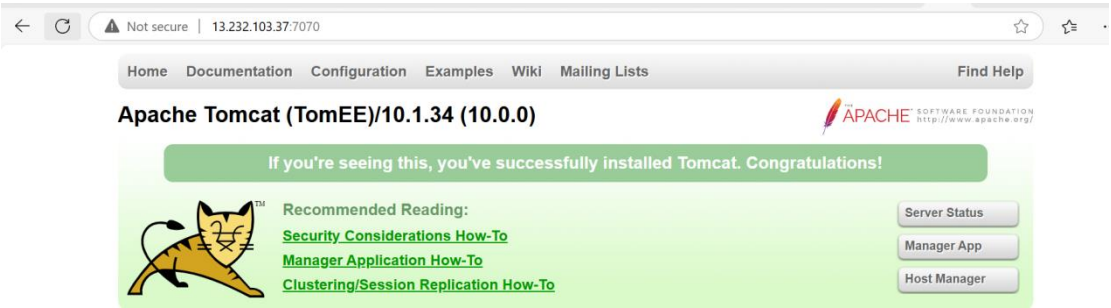
sh install-docker.sh

```
ubuntu@ip-172-31-12-238:~$ sudo su -
root@ip-172-31-12-238:~# curl -fsSL https://get.docker.com -o install-docker.sh
root@ip-172-31-12-238:~# sh install-docker.sh
# Executing docker install script, commit: 4c94a56999e10efcf48c5b8e3f6afea464f9108e
+ sh -c apt-get -qq update >/dev/null
```

TOMCAT INSTALLATION:

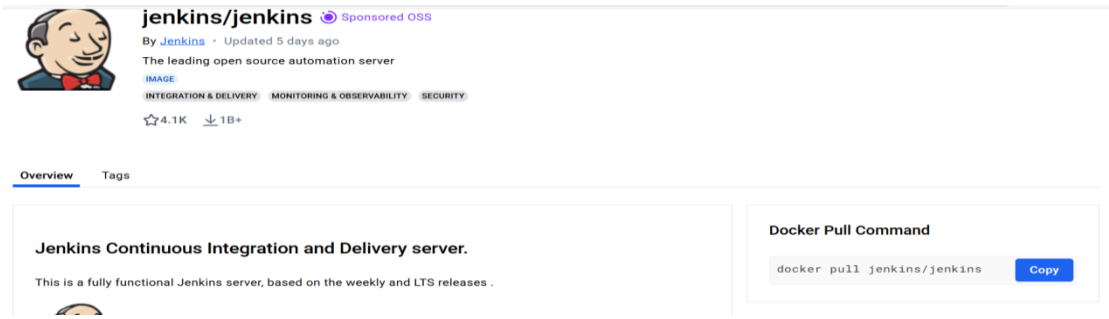
#docker run --name mytomcat -p 7070:8080 tomee
docker run --name mytomcat -p 7070:8080 -d tome

Take the public ip and add 7070 to access tomcat



JENKINS INSTALATION:

docker run --name jenkins -p 9090:8080 jenkins/jenkins



```
root@ip-172-31-12-238:~# docker run --name jenkins -p 9090:8080 jenkins/jenkins
Unable to find image 'jenkins/jenkins:latest' locally
latest: Pulling from jenkins/jenkins
fd0410a2d1ae: Extracting [==========================================>      ]  42.76MB/48.48MB
007307ec8e93: Download complete
1c6f0cfeaa6c: Download complete
d659880150a8: Download complete
8733ebe6e173: Download complete
32deaa29f9f6: Download complete
0dd998eb9cf5: Download complete
0110ea549d15: Download complete
d5cefcd910d4: Download complete
cc0276eb9274: Download complete
67c4d9d41045: Download complete
2810efbbbb3e: Download complete
```

Using docker file:

```
Using docker file
--------------------
This is a simple text file, which uses  predefinied keywords for creating customized docker images.

Key words used in docker file  ( case sensitive )

1) FROM  --  used to specify the base image from which the docker file has to be created.

2) MAINTAINER -- This represents name of the organization or the
 author who created this docker file.

3) CMD   -- This is used to specify the initial command that should be executed when the container
starts.

4) ENTRYPOINT - used to specify the default process that should be executed when container starts.
It can also be used for accepting arguments from the CMD instruction.

5) RUN  -- Used for running linux commands within the container. It is generally helpful for
installing the software in the container.

6) USER  --      used to specify the default user who should login into the container.
```

3) CMD -- This is used to specify the initial command that should be executed when the container starts.

4) ENTRYPOINT - used to specify the default process that should be executed when container starts. It can also be used for accepting arguments from the CMD instruction.

5) RUN -- Used for running linux commands within the container. It is generally helpful for installing the software in the container.

6) USER -- used to specify the default user who should login into the container.

7) WORKDIR --
Used to specify default working directory in the container

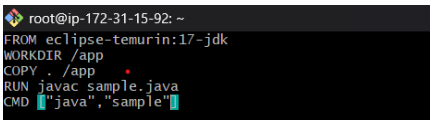
8) COPY -- Copying the files from the host machine to the container.

9) ADD -- Used for copying files from host to container, it can also be used for downloading files from remote servers.

Run: is used to execute commands while building the Docker image**.**
Cmd: defines the default command **that runs** when the container starts**.**

Create dockerfile: **vim dockerfile**

```
FROM eclipse-temurin:17-jdk
WORKDIR /app
COPY . /app
RUN javac sample.java
CMD ["java","sample"]
```



:wq

Create sample.java(calculator program)

vim sample.java

```
import java.util.Scanner;

class sample{
  public static void main(String[] args) {

    char operator;
    Double number1, number2, result;

    // create an object of Scanner class
    Scanner input = new Scanner(System.in);

    // ask users to enter operator
    System.out.println("Choose an operator: +, -, *, or /");
    operator = input.next().charAt(0);

    // ask users to enter numbers
    System.out.println("Enter first number");
    number1 = input.nextDouble();

    System.out.println("Enter second number");
    number2 = input.nextDouble();

    switch (operator) {
```

```
      // performs addition between numbers
      case '+':
        result = number1 + number2;
        System.out.println(number1 + " + " + number2 + " = " + result);
        break;

      // performs subtraction between numbers
      case '-':
        result = number1 - number2;
        System.out.println(number1 + " - " + number2 + " = " + result);
        break;

      // performs multiplication between numbers
      case '*':
        result = number1 * number2;
        System.out.println(number1 + " * " + number2 + " = " + result);
        break;

      // performs division between numbers
      case '/':
        result = number1 / number2;
        System.out.println(number1 + " / " + number2 + " = " + result);
        break;

      default:
        System.out.println("Invalid operator!");
        break;
    }

    input.close();
  }
}
```

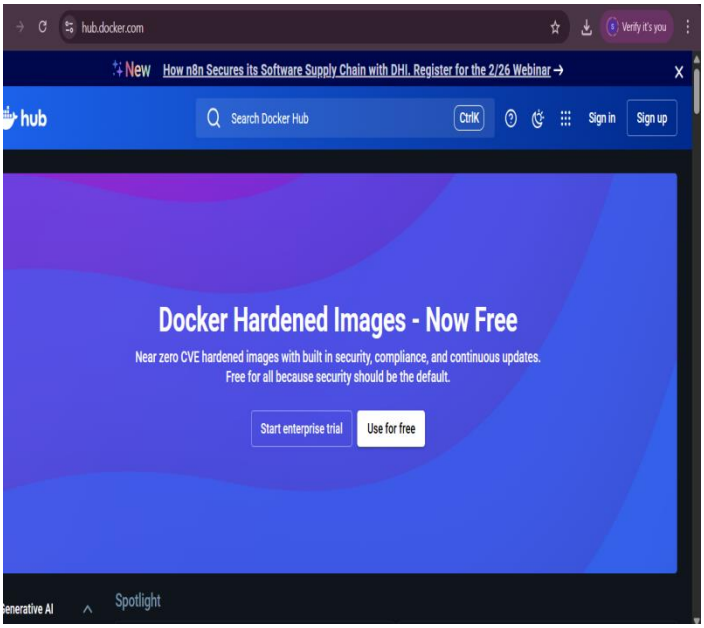
Install java

apt-get install openjdk-17-jdk -y
compile : javac sample.java
run :java sample(sample is the file name)

Compile and run the java program

```
root@ip-172-31-12-238:~# javac sample.java
root@ip-172-31-12-238:~# java sample
Choose an operator: +, -, *, or /
+
Enter first number
10
Enter second number
20
10.0 + 20.0 = 30.0
root@ip-172-31-12-238:~#
```

Create a dockerhub account https://hub.docker.com



Create a docker image for the java application

docker build -t maheedhar45/javacalculator .

```
root@ip-172-31-12-238:~# docker build -t maheedhar45/javacalculator .
[+] Building 16.8s (9/9) FINISHED
 => [internal] load build definition from dockerfile
 => => transferring dockerfile: 119B
 => [internal] load metadata for docker.io/library/openjdk:latest
 => [internal] load .dockerignore
 => => transferring context: 2B
 => [1/4] FROM docker.io/library/openjdk:latest@sha256:9b448de897d211c9e0ec635a485650aed6e28d4eca1efbc34940560a480b3f1f
 => => resolve docker.io/library/openjdk:latest@sha256:9b448de897d211c9e0ec635a485650aed6e28d4eca1efbc34940560a480b3f1f
 => => sha256:9b448de897d211c9e0ec635a485650aed6e28d4eca1efbc34940560a480b3f1f 1.04kB / 1.04kB
 => => sha256:fe05457a5e9b9403f8e72eeba507ae80a4237d2d2d3f219fa62ceb128482a9ee 954B / 954B
 => => sha256:71260f256d19f4ae5c762601e5301418d2516ca591103b1376f063be0b7ba056 4.46kB / 4.46kB
 => => sha256:197c1adcd755131915cd019bdd58658d44445b3638f65449932c18ee39b6047c 44.56MB / 44.56MB
 => => sha256:57b698b7af4b18900b53c768746b1dfb603dfb9aec1eea328fdac86d37001e2a 12.26MB / 12.26MB
 => => sha256:95a27dbe0150755fca4304b4afd0d7d6dd6a40ede6fdb30da8568e9e8cdf23a9 188.74MB / 188.74MB
 => => extracting sha256:197c1adcd755131915cd019bdd58658d44445b3638f65449932c18ee39b6047c
 => => extracting sha256:57b698b7af4b18900b53c768746b1dfb603dfb9aec1eea328fdac86d37001e2a
 => => extracting sha256:95a27dbe0150755fca4304b4afd0d7d6dd6a40ede6fdb30da8568e9e8cdf23a9
 => [internal] load build context
 => => transferring context: 33.36kB
 => [2/4] WORKDIR /app
 => [3/4] COPY . /app
 => [4/4] RUN javac sample.java
 => exporting to image
 => => exporting layers
 => => writing image sha256:4c56e7c240a98376e68ab0d4657190e439631dda16cd5d995f708b51361a29f0
 => => naming to docker.io/maheedhar45/javacalculator
root@ip-172-31-12-238:~#
```

docker image ls

```
root@ip-172-31-12-238:~# docker image ls
REPOSITORY                  TAG       IMAGE ID       CREATED          SIZE
maheedhar45/javacalculator  latest    4c56e7c240a9   53 seconds ago   470MB
mycontainer                 latest    d88c66989850   18 minutes ago   214MB
jenkins/jenkins             latest    f26f1a7cbb33   5 days ago       466MB
tomee                       latest    5567c8574db7   5 weeks ago      367MB
ubuntu                      latest    b1d9df8ab815   2 months ago     78.1MB
root@ip-172-31-12-238:~#
```

Now u push the docker image into docker hub

So create dockerhub account (hub.docker.com) (maheedhar@cmritonline.ac.in) (Welcome@1)
Login

docker login -u maheedhar45(username)
Password: Welcome@1

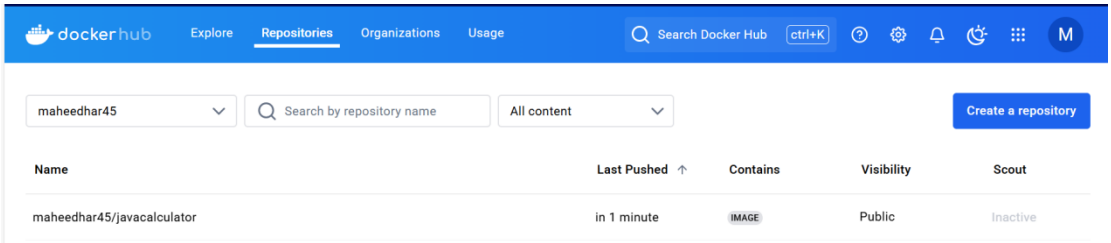
```
root@ip-172-31-12-238:~# docker login -u maheedhar45
Password:
WARNING! Your password will be stored unencrypted in /root/.docker/config.json.
Configure a credential helper to remove this warning. See
https://docs.docker.com/engine/reference/commandline/login/#credential-stores

Login Succeeded
root@ip-172-31-12-238:~#
```

docker push maheedhar45/javacalculator

```
root@ip-172-31-12-238:~# docker push maheedhar45/javacalculator
Using default tag: latest
The push refers to repository [docker.io/maheedhar45/javacalculator]
160ccfe884b2: Pushed
09f939878b71: Pushed
4fc40ef6f284: Pushed
56285d9a7760: Mounted from library/openjdk
077bff59ce57: Mounted from library/openjdk
9cd9df9ffc97: Mounted from library/openjdk
latest: digest: sha256:c2507d7ca91aa526ef92f9af23afc06ba896f8733269702e0b4af25fc074ca7c size: 1577
root@ip-172-31-12-238:~#
```

Now go and check in the docker hub



Accessing the image

docker run --name myjava -it maheedhar45/javacalculator

```
root@ip-172-31-12-238:~# docker run --name myjava -it maheedhar45/javacalculator
Choose an operator: +, -, *, or /
+
Enter first number
10
Enter second number
20
10.0 + 20.0 = 30.0
root@ip-172-31-12-238:~#
```